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Rai Technology University

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CALCRAFT – MODERN CALCULATOR WEB APPLICATION

PROJECT REPORT

Submitted To

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1. Aim of the Project :

The aim of this project is to design and develop a responsive calculator web application using HTML, CSS, and JavaScript. The application performs basic arithmetic operations through an interactive and user-friendly interface. It also demonstrates the practical implementation of front-end web development concepts including HTML structure, CSS styling, JavaScript programming, event handling, and DOM manipulation.

2. Introduction :

Calculators are among the most commonly used applications for performing mathematical computations. Developing a calculator is an excellent way to understand how different web technologies work together to build an interactive application.

The CalCraft project is a responsive calculator developed using HTML5, CSS3, and JavaScript. HTML provides the structure of the application, CSS is responsible for designing an attractive and responsive interface, while JavaScript implements all calculator functionalities such as arithmetic operations, input validation, and keyboard support.

The project provides a modern user interface and works efficiently on both desktop and mobile devices. It also demonstrates the use of event handling, DOM manipulation, functions, variables, operators, conditional statements, and responsive web design.

3. Technologies Used :

Technology	Purpose
HTML5	Structure of the application

CSS3	Styling and responsive design
JavaScript (ES6)	Calculator logic and functionality
Git & GitHub	Version control
Vercel	Deployment

4. Project Description :

CalCraft is a modern web-based calculator application developed using HTML, CSS, and JavaScript. The application provides a clean and responsive interface for performing basic arithmetic calculations.

The calculator supports addition, subtraction, multiplication, division, modulus, decimal calculations, clear, delete, and equal operations. Keyboard shortcuts have also been implemented to improve usability.

Input validation prevents invalid mathematical expressions such as consecutive operators or multiple decimal points within the same number. The project follows a structured folder organization, uses Git and GitHub for version control, and is deployed on Vercel for online access.

5. Features :

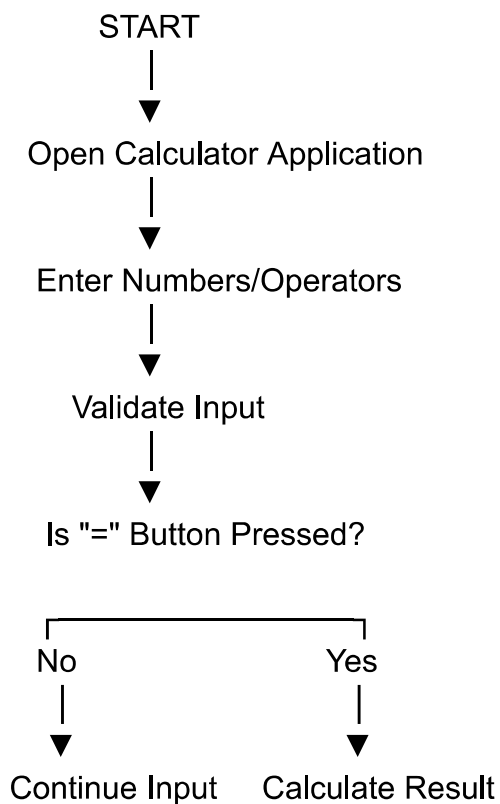
- Modern User Interface
- Responsive Design
- Addition
- Subtraction
- Multiplication
- DivisionModulus (%)
- Decimal Support
- AC (Clear)
- DEL (Delete)

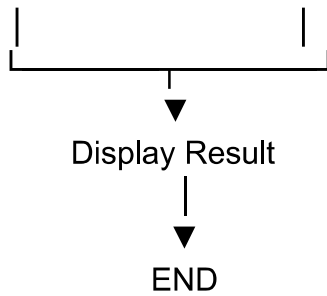
- Keyboard Support
- Input Validation
- GitHub Version Control
- Live Deployment using Vercel
- Developer Credits Footer

6. Algorithm :

1. Start the calculator application.
2. Display the calculator interface.
3. Wait for user input.
4. Accept numbers and operators.
5. Display the entered expression.
6. Validate the entered input.
7. If AC is pressed, clear the display.
8. If DEL is pressed, delete the last entered character.
9. If "=" is pressed, evaluate the expression.
10. Display the calculated result.
11. Repeat until the application is closed.

7. Flowchart :





8. Project Folder Structure :

```
CalCraft/
├── index.html
├── README.md
├── css/
│   └── style.css
├── js/
│   └── script.js
├── assets/
│   ├── images/
│   └── icons/
└── screenshots/
```

9. Source Code :

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Basic Meta Tags -->
```

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<!-- Title -->
<title>CalCraft | Modern Calculator</title>

<!-- CSS -->
<link rel="stylesheet" href="css/style.css">
<link
rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.7.2/css/all.min.css"
/>

<!-- Favicon (Optional) -->
<!-- <link rel="icon" href="assets/icons/favicon.png"> -->
</head>
<body>

<!-- ===== HEADER ===== -->
<header class="header">
  <h1>CalCraft</h1>
</header>

<!-- ===== MAIN ===== -->
<main class="calculator-container">

<section class="calculator">

  <!-- Display -->
  <div class="display">
    <input
      type="text"
      id="display"
      placeholder="0"
      readonly
    >
  </div>

  <!-- Buttons -->
  <div class="buttons">

    <button class="btn function">AC</button>
    <button class="btn function">DEL</button>
    <button class="btn operator">%</button>
```

<button class="btn operator">÷</button>

<button class="btn number">7</button>

<button class="btn number">8</button>

<button class="btn number">9</button>

<button class="btn operator">×</button>

<button class="btn number">4</button>

<button class="btn number">5</button>

<button class="btn number">6</button>

<button class="btn operator">-</button>

<button class="btn number">1</button>

<button class="btn number">2</button>

<button class="btn number">3</button>

<button class="btn operator">+</button>

<button class="btn number">0</button>

<button class="btn number">.</button>

<button class="btn equal">=</button>

</div>

</section>

</main>

<!-- ===== FOOTER ===== -->

<!-- ===== FOOTER ===== -->

<footer class="footer">

<h2>Meet the Developers</h2>

<div class="team-container">

<!-- Mohammed Sahil -->

<div class="team-member">

<h3>Mohammed Sahil</h3>

<div class="social-links">


```
        <i class="fab fa-github"></i>
    </a>

    <a href="https://www.linkedin.com/in/mdsahil01" target="_blank"
rel="noopener noreferrer" aria-label="Mohammed Sahil LinkedIn">
        <i class="fab fa-linkedin"></i>
    </a>
</div>
</div>

<!-- Pavan Vastrad -->
<div class="team-member">
    

    <h3>Pavan Vastrad</h3>

    <div class="social-links">
        <a href="https://github.com/vastradpavan88" target="_blank"
rel="noopener noreferrer" aria-label="Pavan Vastrad GitHub">
            <i class="fab fa-github"></i>
        </a>

        <a href="https://www.linkedin.com/in/pavan-vastrad/" target="_blank"
rel="noopener noreferrer" aria-label="Pavan Vastrad LinkedIn">
            <i class="fab fa-linkedin"></i>
        </a>
    </div>
</div>

</div>

<p class="copyright">
    © 2026 CalCraft. Developed by Mohammed Sahil & Pavan Vastrad.
</p>

</footer>

<!-- JavaScript -->
<script src="js/script.js"></script>

</body>
</html>
```


CSS

```
/* =====
   Google Font
   ===== */
@import
url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600&displ
ay=swap');

/* =====
   CSS Reset
   ===== */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

/* =====
   Variables
   ===== */
:root {
  --bg-color: #0f172a;
  --card-color: #1e293b;
  --text-color: #ffffff;
  --shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
  --radius: 20px;
}

/* =====
   Body
   ===== */
body {
  font-family: 'Poppins', sans-serif;
  background: linear-gradient(135deg, #0f172a, #1e3a8a);
  color: var(--text-color);

  min-height: 100vh;

  display: flex;
  flex-direction: column;
}
```

```

/* =====
Header
===== */
.header {
  text-align: center;
  padding: 25px 0;
}

.header h1 {
  font-size: 2rem;
  font-weight: 600;
}

/* =====
Main
===== */
.calculator-container {
  flex: 1;

  display: flex;
  justify-content: center;
  align-items: center;

  padding: 20px;
}

/* =====
Calculator Card
===== */
.calculator {
  width: 360px;
  background: var(--card-color);
  border-radius: var(--radius);
  padding: 25px;
  box-shadow: var(--shadow);
}

/* =====
Footer
===== */

.footer {
  background: #0f172a;

```

```
    color: #ffffff;
    text-align: center;
    padding: 50px 20px;
    margin-top: 50px;
}
```

```
.footer h2 {
    margin-bottom: 30px;
    font-size: 2rem;
}
```

```
.team-container {
    display: flex;
    justify-content: center;
    gap: 40px;
    flex-wrap: wrap;
}
```

```
.team-member {
    background: #1e293b;
    padding: 20px;
    border-radius: 15px;
    width: 250px;
    transition: transform 0.3s ease, box-shadow 0.3s ease;
}
```

```
.team-member:hover {
    transform: translateY(-8px);
    box-shadow: 0 10px 20px rgba(0,0,0,0.3);
}
```

```
.team-member img {
    width: 120px;
    height: 120px;
    object-fit: cover;
    border-radius: 50%;
    border: 4px solid #3b82f6;
    margin-bottom: 15px;
}
```

```
.team-member h3 {
    margin-bottom: 15px;
}
```

```
.social-links {  
  display: flex;  
  justify-content: center;  
  gap: 20px;  
}
```

```
.social-links a {  
  color: #ffffff;  
  font-size: 1.8rem;  
  transition: color 0.3s ease, transform 0.3s ease;  
}
```

```
.social-links a:hover {  
  color: #3b82f6;  
  transform: scale(1.2);  
}
```

```
.copyright {  
  margin-top: 30px;  
  font-size: 0.95rem;  
  opacity: 0.8;  
}
```

```
@media (max-width: 768px) {  
  .team-container {  
    flex-direction: column;  
    align-items: center;  
  }
```

```
  .team-member {  
    width: 90%;  
    max-width: 320px;  
  }  
}
```

```
/* =====  
Display  
===== */
```

```
.display {  
  margin-bottom: 20px;  
}
```

```
#display {  
  width: 100%;  
  height: 70px;
```

```

border: none;
outline: none;
border-radius: 12px;
background-color: #0f172a;
color: #fff;
font-size: 2rem;
text-align: right;
padding: 0 20px;
}

/* =====
   Button Grid
===== */
.buttons {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 15px;
}

/* =====
   Base Button Style
===== */
.btn {
  height: 65px;
  border: none;
  border-radius: 12px;
  font-size: 1.2rem;
  font-weight: 600;
  cursor: pointer;
}

/* =====
   Number Buttons
===== */
.number {
  background-color: #334155;
  color: white;
}

/* =====
   Operator Buttons
===== */
.operator {
  background-color: #2563eb;

```

```

    color: white;
}

/* =====
   Function Buttons
===== */
.function {
    background-color: #ef4444;
    color: white;
}

/* =====
   Equal Button
===== */
.equal {
    background-color: #22c55e;
    color: white;
}

/* =====
   Button Hover Effects
===== */
.btn {
    transition: all 0.2s ease;
}

.btn:hover {
    transform: translateY(-2px);
    filter: brightness(1.1);
}

.btn:active {
    transform: scale(0.96);
}

/* =====
   Display Focus
===== */
#display {
    box-shadow: inset 0 0 8px rgba(255, 255, 255, 0.1);
}

```

```

/* =====
Calculator Animation
===== */
.calculator {
  animation: fadeIn 0.6s ease-in-out;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(20px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

/* =====
Responsive Design
===== */
@media (max-width: 480px) {
  .calculator {
    width: 100%;
    padding: 20px;
  }

  .btn {
    height: 55px;
    font-size: 1rem;
  }

  #display {
    height: 60px;
    font-size: 1.6rem;
  }

  .header h1 {
    font-size: 1.6rem;
  }
}

```

JavaScript

```
// =====  
// Select Elements  
// =====  
  
const display = document.getElementById("display");  
  
const numberButtons = document.querySelectorAll(".number");  
const operatorButtons = document.querySelectorAll(".operator");  
const functionButtons = document.querySelectorAll(".function");  
const equalButton = document.querySelector(".equal");  
  
// =====  
// Variables  
// =====  
  
let currentInput = "";  
  
// =====  
// Helper Functions  
// =====  
  
function updateDisplay() {  
    display.value = currentInput;  
}  
  
function isOperator(value) {  
    return ["+", "-", "*", "/", "%"].includes(value);  
}  
  
// =====  
// Number Button Events  
// =====  
  
numberButtons.forEach(button => {  
    button.addEventListener("click", () => {  
  
        if (button.textContent === ".") {  
            const parts = currentInput.split(/[+\-*/%/]);
```



```

        if (parts[parts.length - 1].includes(".")) {
            return;
        }
    }

    currentInput += button.textContent;
    updateDisplay();
});
});

// =====
// Operator Button Events
// =====

operatorButtons.forEach(button => {
    button.addEventListener("click", () => {

        const operator = button.textContent;

        // Prevent starting with an operator
        if (currentInput === "") {
            return;
        }

        const lastChar = currentInput.slice(-1);

        // Prevent consecutive operators
        if (
            isOperator(lastChar) &&
            isOperator(operator === "x" ? "*" : operator === "÷" ? "/" : operator)
        ) {
            return;
        }

        if (operator === "x") {
            currentInput += "*";
        } else if (operator === "÷") {
            currentInput += "/";
        } else {
            currentInput += operator;
        }

        updateDisplay();
    });
});

```

```

    });
  });

  // =====
  // Function Buttons
  // =====

  functionButtons.forEach(button => {
    button.addEventListener("click", () => {

      const action = button.textContent;

      if (action === "AC") {
        currentInput = "";
      }

      if (action === "DEL") {
        currentInput = currentInput.slice(0, -1);
      }

      updateDisplay();
    });
  });

  // =====
  // Equal Button
  // =====

  equalButton.addEventListener("click", () => {

    if (currentInput.trim() === "") {
      return;
    }

    try {
      currentInput = eval(currentInput.toString());
      updateDisplay();
    } catch {
      display.value = "Error";

      setTimeout(() => {
        currentInput = "";
        updateDisplay();
      }, 1000);
    }
  });

```

```

    }
  });

  // =====
  // Keyboard Support
  // =====

  document.addEventListener("keydown", (event) => {

    const key = event.key;

    const allowedKeys = [
      "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
      ".", "+", "-", "*", "/", "%",
      "Enter", "Backspace", "Escape"
    ];

    if (!allowedKeys.includes(key)) {
      return;
    }

    const lastChar = currentInput.slice(-1);

    // Prevent starting with an operator
    if (
      currentInput === "" &&
      ["+", "-", "*", "/", "%"].includes(key)
    ) {
      return;
    }

    // Prevent consecutive operators
    if (
      ["+", "-", "*", "/", "%"].includes(key) &&
      ["+", "-", "*", "/", "%"].includes(lastChar)
    ) {
      return;
    }

    // Prevent multiple decimal points
    if (key === ".") {
      const parts = currentInput.split(/[\+\-\*\/%]/);

      if (parts[parts.length - 1].includes(".")) {

```

```

        return;
    }
}

// Numbers & Operators
if (
    (key >= "0" && key <= "9") ||
    key === "." ||
    key === "+" ||
    key === "-" ||
    key === "*" ||
    key === "/" ||
    key === "%"
) {
    currentInput += key;
    updateDisplay();
}

// Calculate
if (key === "Enter") {
    equalButton.click();
}

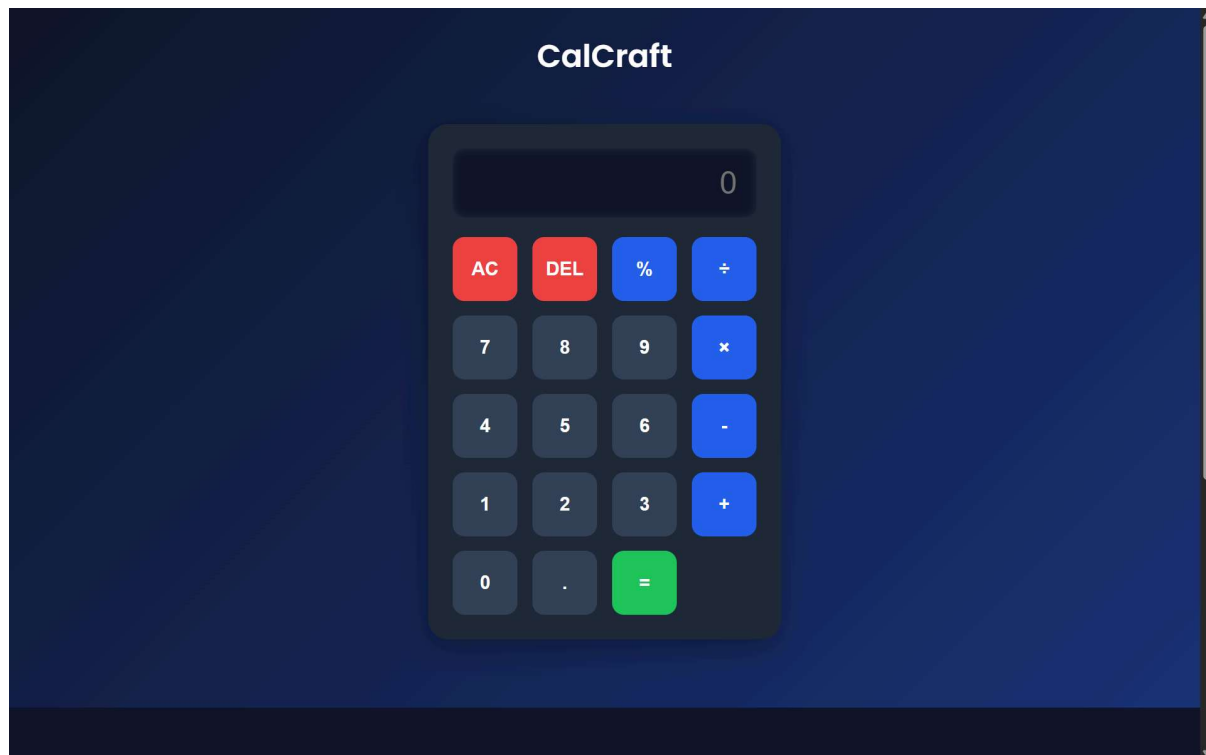
// Delete
if (key === "Backspace") {
    currentInput = currentInput.slice(0, -1);
    updateDisplay();
}

// Clear
if (key === "Escape") {
    currentInput = "";
    updateDisplay();
}
});

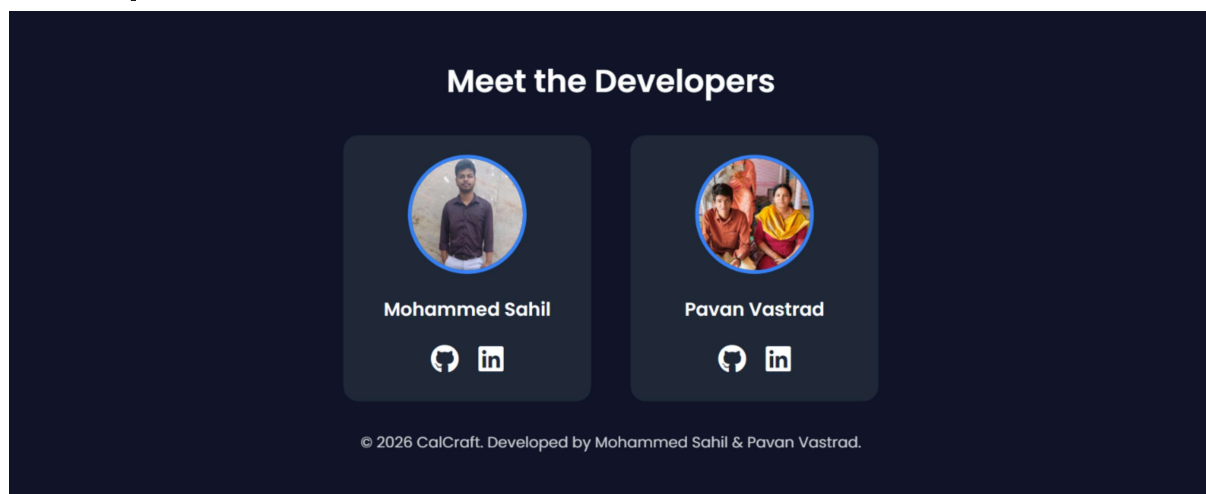
```

10. Output Screenshots :

Calculator Home Screen :



Developer Credits Footer :



GitHub Repository :

<https://github.com/Mdsahil01/cal-craft>

Live Vercel Website :

<https://cal-craftt.vercel.app/>

11. Test Cases :

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC-01	Addition	$5 + 5 =$	10	10	Pass
TC-02	Subtraction	$10 - 4 =$	6	6	Pass
TC-03	Multiplication	$6 \times 7 =$	42	42	Pass
TC-04	Division	$20 \div 5 =$	4	4	Pass
TC-05	Modulus	$10 \% 3 =$	1	1	Pass
TC-06	Decimal Calculation	$3.5 + 2.5 =$	6	6	Pass

TC-07	Clear Button (AC)	AC	Display Cleared	Display Cleared	Pass
TC-08	Delete Button (DEL)	123 DEL	12	12	Pass
TC-09	Keyboard Support	5 + 2 + Enter	7	7	Pass
TC-10	Consecutive Operators	5++6	Prevent Invalid Input	Prevented	Pass
TC-11	Multiple Decimal Points	3.5.2	Prevent Invalid Input	Prevented	Pass
TC-12	Empty Expression	Press "=" on Empty Display	No Calculation	No Calculation	Pass
TC-13	Responsive Layout	Open on Mobile Device	Responsive UI	Responsive UI	Pass
TC-14	GitHub & LinkedIn Links	Click Footer Icons	Open Correct Profile	Opened Successfully	Pass

TC-15	Live Deployment	Open Vercel URL	Calculator Loads Successfully	Working	Pass
-------	-----------------	-----------------	-------------------------------	---------	------

12. Conclusion :

The CalCraft project successfully demonstrates the development of a responsive calculator using HTML, CSS, and JavaScript. The application performs all basic arithmetic operations accurately while maintaining a clean and user-friendly interface.

Through this project, practical knowledge of HTML structure, CSS styling, JavaScript programming, DOM manipulation, event handling, responsive web design, GitHub version control, and Vercel deployment was gained. The project fulfills the objectives of the assignment and provides a strong foundation in front-end web development.

13. Future Enhancements :

- Scientific Calculator Functions
- Dark Mode
- Light Mode
- Calculation History
- Memory Functions
- Voice Input
- Theme Customization
- Copy Result Button
- Unit Converter

14. References :

<https://developer.mozilla.org/>

<https://www.w3schools.com/>

<https://fontawesome.com/>

<https://github.com/>

<https://vercel.com/>